

STRENGTHENING GRANTS DATA SYSTEM

US Department of Agriculture | Local and Regional Foods Division Research Branch

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Keywords:

data validation, data pipeline, web scraping, ETL, document parsing

Summary:

To make the USDA AMS Grant Division's data more robust, Jackie documented the main repository behind 80% of the division's grant data requests, and built the foundation for a data validation layer. He diagrammed the repository's scripts and data sources, ran a dependency audit that reduced the environment from nearly 200 packages to the dozen actually in use, and built the data collection stage of a validation layer in **Python** - an automated process that webscrapes AMS grant pages, detects reports in Excel and PDF formats, downloads, and standardizes the report data. This work strengthens the documentation for the division's most-used data, and lays the groundwork for data validation to strengthen the accuracy of the data.

PROGRAM EVALUATION ANALYSIS

US Department of Agriculture | Local and Regional Foods Division Research Branch

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Keywords:

data cleaning, natural language processing, TF-IDF

Summary:

To make grantee-reported results comparable across the division's grant programs, Jackie tested the crosswalk that collapses more than 1,800 unique program-specific evaluation questions (“outcome-indicators”) into a shared set of ~300 outcome-indicators. Using **Python**, he built a reusable framework for applying the crosswalk to any program, diagnosed unmapped indicators, and benchmarked automated matching techniques against the staff-verified mapping. The result is a framework the division can reuse rather than rebuild, and evidence that **scikit-learn's TF-IDF** can reduce 90% of future mapping work.

The Local and Regional Foods Division (LRFD) supports local food producers, markets, and communities by providing easy access to the **ideas, innovations and research** necessary to grow and sustain a productive business.

Local Food Research and Development

Regional Food Business Centers

Local and Regional Food Systems Research and Publications

Local Food Directories

Food Facility Design

The Local and Regional Foods Division (LRFD) has three main roles as a researcher, a convener, and a technical assistance provider. In the U.S. Department of Agriculture's (USDA) vision, local food producers, markets, and communities have easy access to ideas, innovations, and research in order to grow and sustain a productive business. This information ensures that opportunities for U.S. food producers are readily available and communities are equipped to successfully grow and sell regionally produced foods, while also supporting increased access to locally produced foods.

The [USDA Local and Regional Food Systems Resource Guide \(pdf\)](#) offers a comprehensive summary of USDA programs that support local and regional food systems.

STRENGTHENING GRANTS DATA SYSTEM

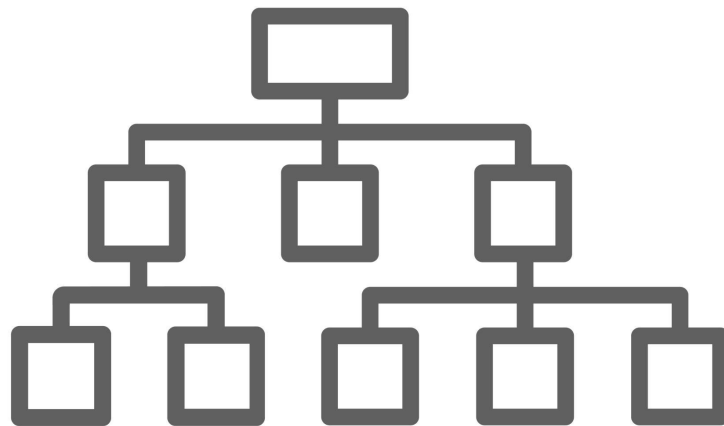
For the AMS Grants Division central data repository

80%

Grant Division's data requests rely on this one system

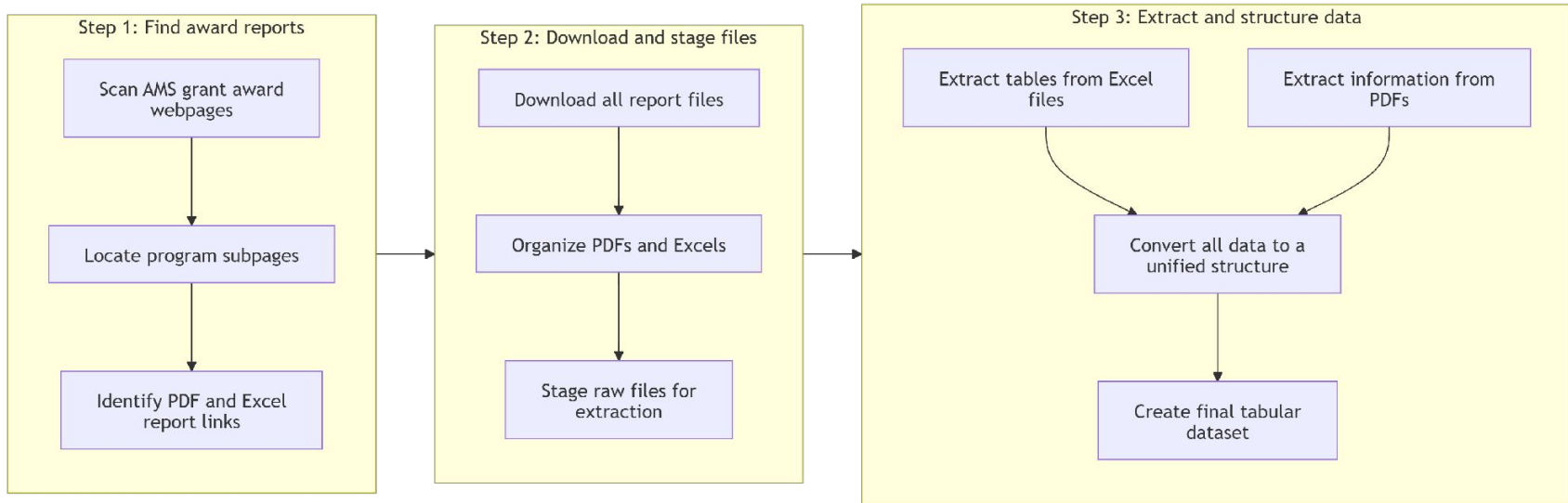
MAPPING THE GRANTS DIVISION DATA

- Anyone can now trace how data moves through the repository, because the tables, scripts, and dependencies between them are diagrammed for the first time.
- The environment runs on 12 packages instead of ~200, after a full dependency audit.



BUILDING DATA VALIDATION LAYER

Diagram of the automated data pipeline



This system sources 70+ reports from the AMS Grant Awards pages that can be used to validate the Grants Division's data

PROGRAM EVALUATION ANALYSIS

Analyzing outcomes and indicators across 12+ AMS programs

1,800+

Unique “outcome-indicators” across all AMS Grant Programs

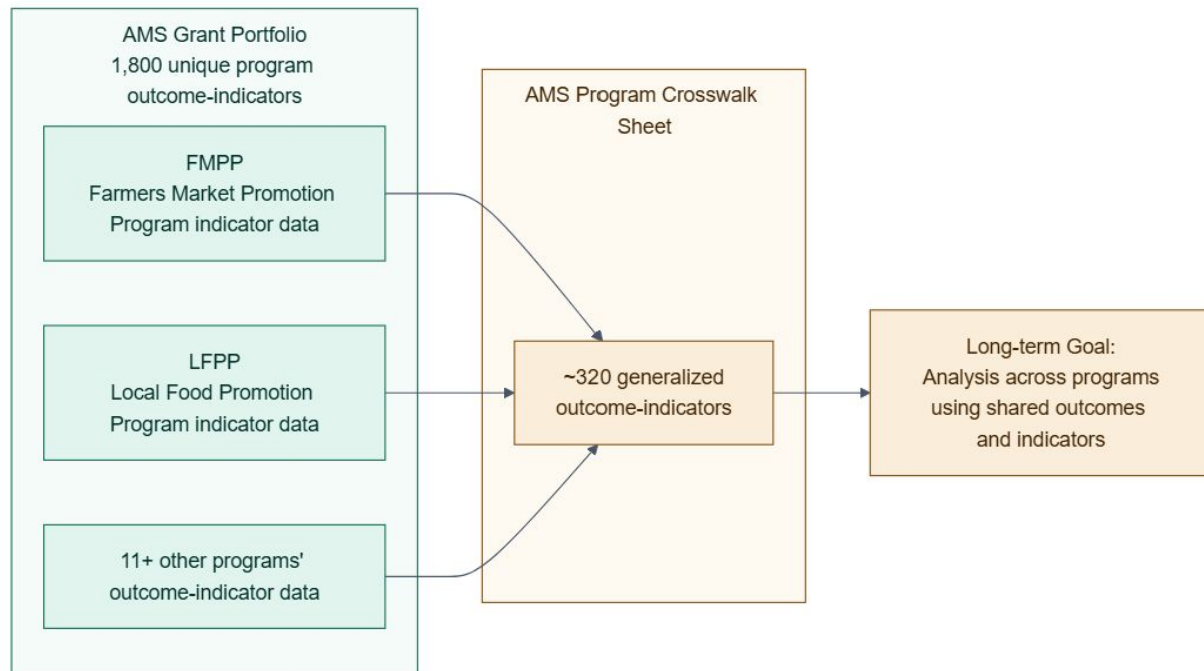
OUTCOMES AND INDICATORS DATA

- Agricultural Marketing Services (AMS) grant awardees complete program evaluations that measure **outcomes** across various **indicators** (i.e., questions) that were developed for their specific program.
- AMS runs 12+ programs, and each program has its own set of indicators.
- Result: over 1,800 uniquely worded outcomes and indicators.

Comparing results across programs requires a shared vocabulary.

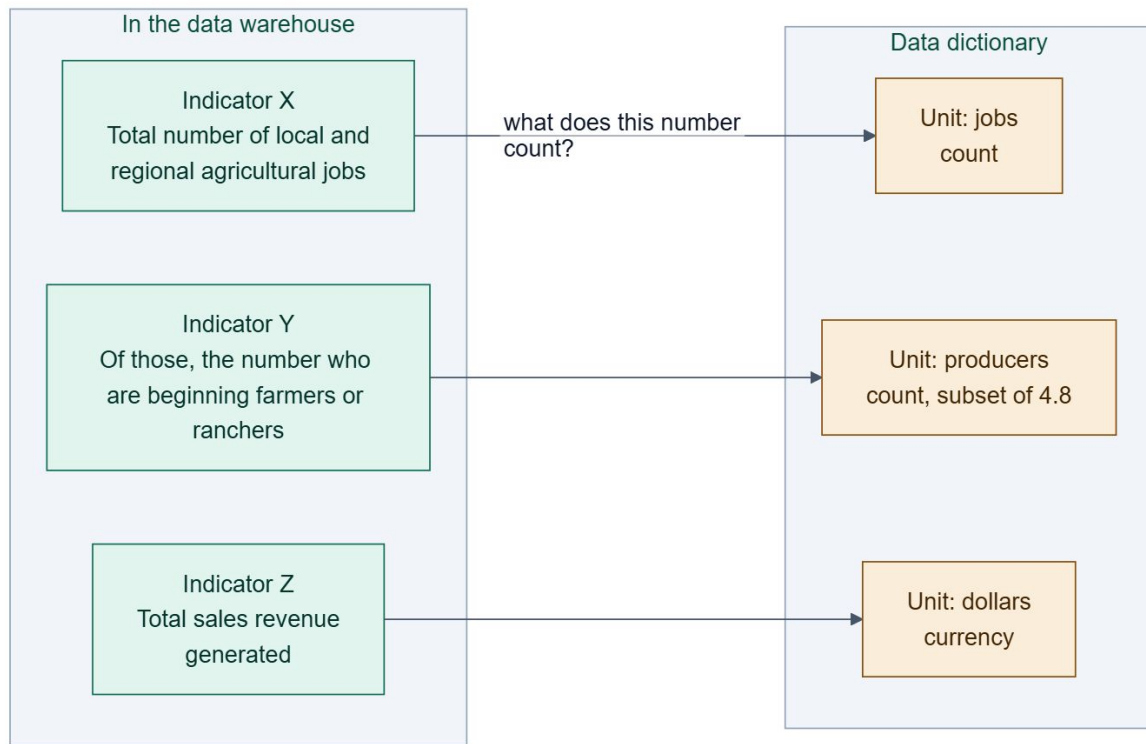
PROGRAM CROSSWALK FILE

Program indicator data can now be bridged with the program crosswalk sheet, allowing all 13 programs to be analyzed together under a “shared vocabulary”



DATA DICTIONARY

Program indicator data now is labeled with the value units, so no one has to ask a program analyst what a number means before they can use it





Thank you!

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